

Technical Report: Latent Dirichlet Allocation (LDA) for Customer Sentiment Analysis

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Course: Data Science & Machine Learning

Project: Topic Modelling of McDonald's Store Reviews Using Latent Dirichlet Allocation (LDA)

1. Executive Summary

This project applies Latent Dirichlet Allocation (LDA), an unsupervised machine learning algorithm, to identify the main themes discussed in 33,396 customer reviews from McDonald's stores. Instead of manually reading thousands of reviews, LDA automatically discovers hidden topics based on word co-occurrence patterns. The analysis identified six major customer experience themes covering service efficiency, food quality, order accuracy, restaurant operations, dining environment, and customer service. These insights provide valuable evidence for improving customer satisfaction and operational performance.

2. Methodology

Exploratory Data Analysis (EDA): The dataset was explored to determine the total number of reviews, identify missing values, and examine the distribution of customer ratings. Visualizations of rating distribution and review length were created.

Text Preprocessing: Reviews were converted to lowercase, punctuation and numbers removed, stop words removed, text tokenized, and words lemmatized to improve topic quality.

LDA Topic Modelling: An LDA model was built using Gensim. Several topic numbers were evaluated, and six topics were selected based on coherence score because they provided the best balance between model quality and interpretability.

3. Topic Modelling Results

Wait Times & Drive-Thru Service (4.43) – Highly Positive

Positive Service & Food Quality (3.79) – Positive

Store Hours & Operations (3.77) – Neutral-Positive

Order Accuracy Issues (3.27) – Neutral

Food & Drink Freshness (2.24) – Negative

Dine-In Experience & Ambience (2.20) – Highly Negative

4. Discussion

Positive reviews emphasized efficient drive-thru service, friendly staff, and good food quality. Negative reviews focused on cold food, cleanliness, poor dining environments, and staff interactions. The LDA model successfully transformed thousands of reviews into actionable business insights.

5. Business Recommendations

1. Improve dine-in facilities and cleanliness.
2. Strengthen food quality control and temperature monitoring.
3. Replicate successful drive-thru practices across stores.
4. Improve order accuracy through verification procedures.
5. Ensure consistent store operations and staffing.

6. Conclusion

The LDA model effectively extracted meaningful themes from large-scale customer reviews. The findings provide evidence-based insights that can help McDonald's improve customer satisfaction and operational performance.